08 · Glaucoma and cataracts

The words of this track, as it says them. Bold brown is the examiner.

Glaucoma and cataracts. Four marks.

Explain glaucoma and cataracts: which part of the eye each one affects, what goes wrong, and how it affects vision.

Another one from her help board, and she sums them up in a few words each. Glaucoma: the pressure in the eye. Cataract: the clarity of the lens.

The hook: glaucoma is a pressure problem, because the drain blocks. A cataract is a clarity problem, because the lens goes cloudy.

Glaucoma first. It involves the aqueous humour, the watery fluid at the front of the eye. Normally the ciliary body makes it and the canal of Schlemm drains it, at the same rate. If the canal of Schlemm is blocked, or too much fluid is made, aqueous humour builds up in the anterior segment, and the intraocular pressure rises. That pressure compresses the retina and its blood supply, and damages the optic nerve. So vision is lost, and it can end in blindness.

Now the cataract. A cataract affects the lens. The proteins in the lens denature, so the lens turns cloudy, or opaque, instead of clear. The cloudy lens stops light reaching the retina properly, so vision becomes blurred and yellowed, with poor night vision and sensitivity to glare. It's common with age and with diabetes. And it's treated with surgery that replaces the lens.

The traps. Don't swap them. Glaucoma is pressure: the aqueous humour and the canal of Schlemm. A cataract is the lens. Second: the ciliary body makes the fluid, and the canal of Schlemm drains it. Third: in glaucoma the lens is fine. It's the retina and the optic nerve that get damaged.

Once more. Glaucoma and cataracts.

Glaucoma: aqueous humour is made by the ciliary body and drained by the canal of Schlemm. If the drain blocks, fluid builds up, the intraocular pressure rises, and it damages the retina and the optic nerve, which can end in blindness. A cataract: the proteins of the lens denature, so it goes cloudy. Vision becomes blurred and yellowed, with glare. It's treated with surgery that replaces the lens.